BEYOND MULTIPLE LINEAR REGRESSION · WEEK 2

When treatment effects vary

Interactions, random slopes and crossed random effects

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METHODS · DESIGN · STATISTICS

Week 1 recap: comparing the models

THE REHABILITATION TRIAL

Simulated trial: 120 patients after knee replacement at six centers.

Within each center: 10 patients randomized to each treatment.

Standard care: weekly physiotherapy and home exercises.

Enhanced care: standard care plus a weekly supervised session.

Outcome: six-minute walk distance at 12 weeks.

THE TREATMENT COMPARISON

Model	Estimate (m)	Estimated SE (m)
Treatment only	11.0	5.77
Fixed center effects	11.0	4.34
Random center effects	11.0	4.34

All three models assumed a common treatment effect across centers.

Fixed center parameters and a random center distribution gave the same treatment estimate and estimated standard error in this setting.

Changing treatment allocation within centers

ORIGINAL ALLOCATION

Center	Standard care	Enhanced care
A	10	10
B	10	10
C	10	10
D	10	10
E	10	10
F	10	10
Total	60	60

UNEQUAL ALLOCATION

Center	Standard care	Enhanced care
A	5	15
B	8	12
C	10	10
D	10	10
E	12	8
F	15	5
Total	60	60

Only allocation changes: the underlying treatment effect remains 10 m in every center.

The models now give different treatment estimates

Unequal allocation within centers; estimated enhanced-minus-standard difference in walk distance.

Model	Estimate (m)	Estimated SE (m)
Treatment only	3.87	5.75
Fixed center effects	8.40	4.56
Random center effects	8.13	4.55

Accounting for center changes the estimate substantially. The two center models give slightly different estimates, with almost identical estimated standard errors.

The models use different comparisons

TREATMENT ONLY

The model compares the average outcome of all enhanced-care patients with that of all standard-care patients.

Unequal allocation gives the groups different mixtures of centers. Their difference reflects both treatment and differences in center baselines.

FIXED CENTER EFFECTS

The model allows each center to have its own baseline outcome.

It estimates treatment by combining comparisons between enhanced and standard care within the same center.

These comparisons remove the shared center baseline.

RANDOM CENTER EFFECTS

The model uses comparisons both within and between centers.

Do centers with more enhanced care have higher average outcomes?

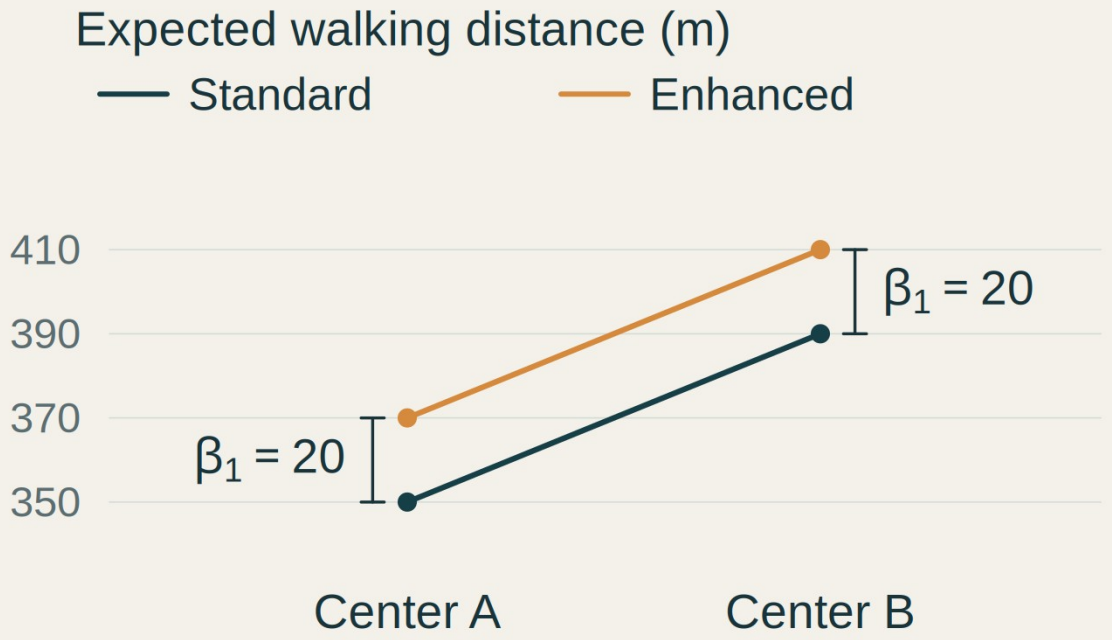
Using between-center information assumes baseline center effects are unrelated to treatment allocation.

So far, we have assumed the same treatment effect in every center. What changes when the treatment effect varies across centers?

A common treatment effect in both centers

Hypothetical example: assume the four true group means are known.

Center	Standard	Enhanced	Row average
A	$\mu_{0A} = 350$	$\mu_{1A} = 370$	**360**
B	$\mu_{0B} = 390$	$\mu_{1B} = 410$	**400**
Column average	$\beta_0 = 370$	390	**380**



μ_{tj}: expected outcome for treatment t in center j

$$Y_{ij} = \underbrace{\beta_0 + c_j}_{\text{Standard-care mean in center } j} + \underbrace{\beta_1 T_{ij}}_{\text{Treatment contribution}} + \underbrace{\varepsilon_{ij}}_{\text{Individual deviation}}$$

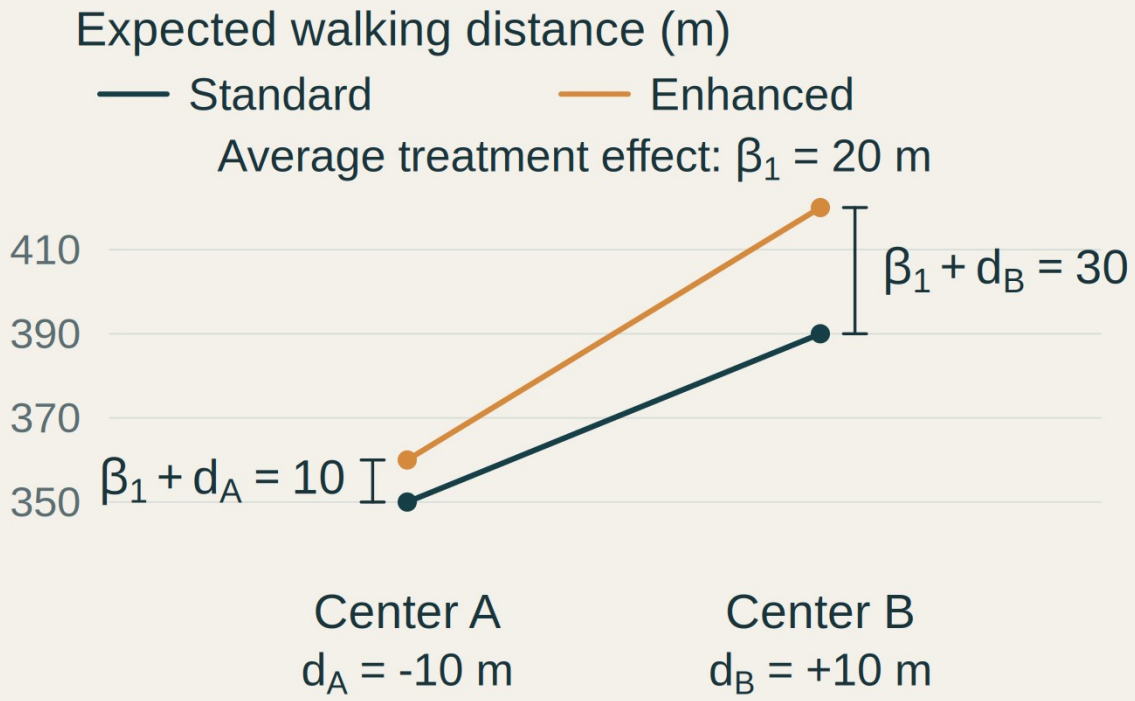
T_{ij} = 0 for standard care and 1 for enhanced care. i = patient; j = center.

Individual patient outcomes vary around their group mean through ε_{ij}.

Different treatment effects across centers

Hypothetical example: assume the four true group means are known.

Center	Standard	Enhanced	Row average
A	$\mu_{0A} = 350$	$\mu_{1A} = 360$	**355**
B	$\mu_{0B} = 390$	$\mu_{1B} = 420$	**405**
Column average	$\beta_0 = 370$	390	**380**



μ_{tj}: expected outcome for treatment t in center j

$$Y_{ij} = \underbrace{\beta_0 + c_j}_{\text{Standard-care mean in center } j} + \underbrace{(\beta_1 + d_j)T_{ij}}_{\text{Treatment contribution}} + \underbrace{\varepsilon_{ij}}_{\text{Individual deviation}}$$

T_{ij} = 0 for standard care and 1 for enhanced care. i = patient; j = center.

Individual patient outcomes vary around their group mean through ε_{ij}.

How should we represent differences between centers?

Beyond two centers, we still allow differences in baseline outcome and treatment effect.

FIXED CENTER EFFECTS

Give each included center its own baseline deviation and treatment-effect deviation.

Express the average treatment effect across the included centers.

RANDOM CENTER EFFECTS

Describe baseline deviations and treatment-effect deviations through a population distribution.

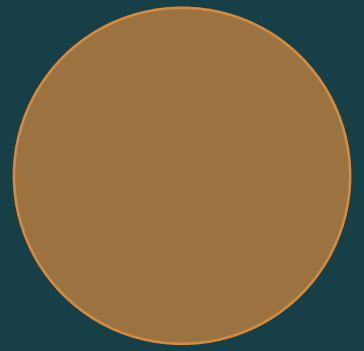
Express the average treatment effect in the population represented by those centers.

We start with fixed center effects and ask what we can learn about treatment in the included centers.

PART 1

Treatment-by-center interaction: fixed effects





Case study: CPAP and blood pressure

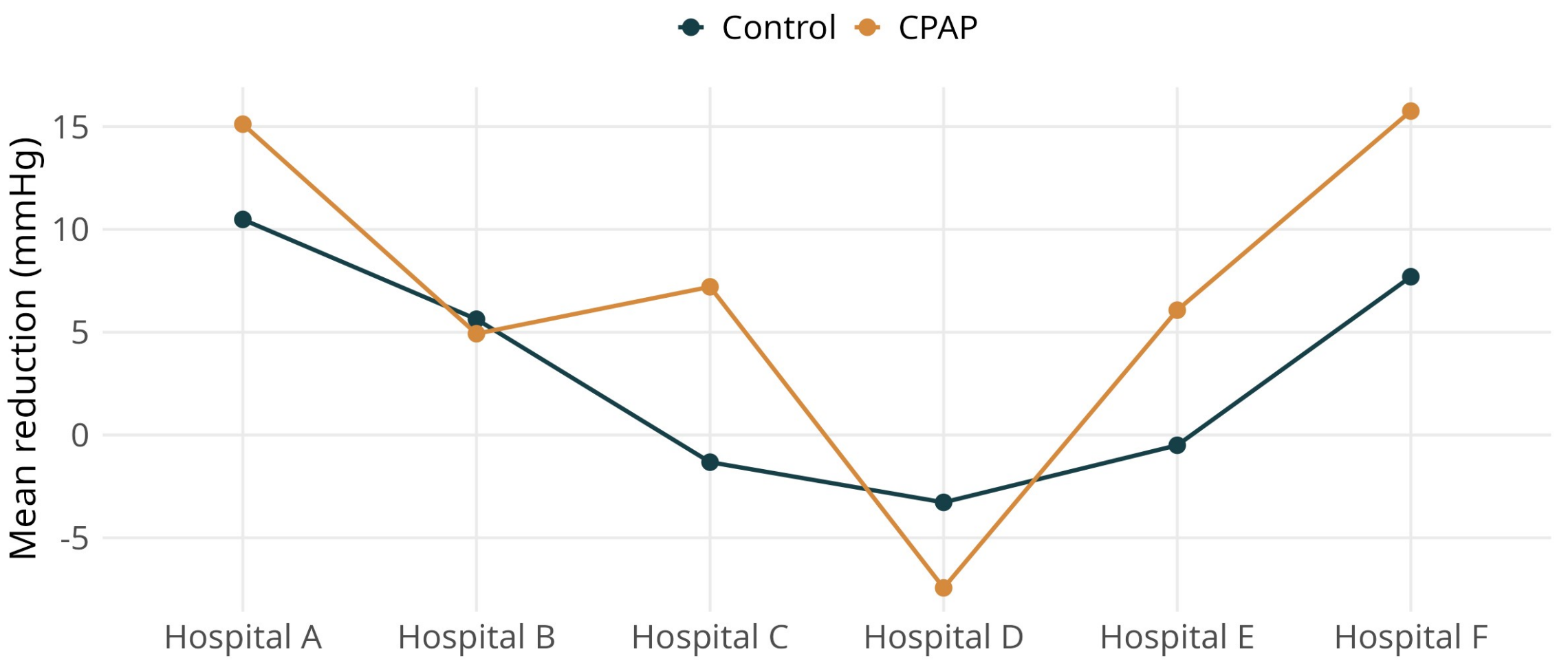
A SIMULATED MULTICENTER TRIAL

- 240 patients with sleep apnea and hypertension despite medication, across six hospitals.
- Hospital sizes: 24, 32, 36, 44, 48 and 56; equal treatment allocation within each.
- Control: continue usual blood-pressure medication.
- CPAP: add continuous positive airway pressure through a mask during sleep.
- Outcome: reduction in 24-hour mean blood pressure after 12 weeks (mmHg).

Positive values indicate a blood-pressure reduction. Inspired by HIPARCO (Martínez-García et al., JAMA, 2013); simulated patient data.



Blood-pressure reduction by treatment and hospital



A fixed-effects model for treatment and center

$$Y_{ij} = \beta_0 + c_j + (\beta_1 + d_j)T_{ij} + \varepsilon_{ij}$$

Y_{ij}: blood-pressure reduction (mmHg). T_{ij} = 0 for control and 1 for CPAP.

CONTROL OUTCOMES

β_0: average control-group reduction across the included centers.

c_j: center j's deviation from that control average.

$$\sum_j c_j = 0$$

TREATMENT EFFECTS

β_1: average treatment effect across the included centers.

d_j: center j's deviation from the average treatment effect.

$$\sum_j d_j = 0$$

Each center receives equal weight in the averages. Center deviations are fixed, unknown parameters.

Fixed-effects interaction model: results

Term	Estimate (mmHg)	Estimated SE (mmHg)	95% CI (mmHg)	p-value
Intercept (β_0)	3.1	0.75	1.6 to 4.6	< .001
CPAP (β_1)	3.8	1.06	1.7 to 5.9	< .001
Hospital A	7.4	2.00	3.4 to 11.3	< .001
Hospital B	2.5	1.77	-1.0 to 6.0	0.157
Hospital C	-4.4	1.69	-7.8 to -1.1	0.009
Hospital D	-6.4	1.56	-9.5 to -3.3	< .001
Hospital E	-3.6	1.51	-6.6 to -0.6	0.017
CPAP × Hospital A	0.8	2.82	-4.8 to 6.4	0.774
CPAP × Hospital B	-4.5	2.50	-9.5 to 0.4	0.072
CPAP × Hospital C	4.7	2.39	0.0 to 9.4	0.050
CPAP × Hospital D	-8.0	2.20	-12.3 to -3.6	< .001
CPAP × Hospital E	2.8	2.13	-1.5 to 7.0	0.198

Effects coding: Hospital F follows from the sum-to-zero constraints.

Which treatment effect do we want to estimate?

SIMPLE TREATMENT EFFECTS

How do treatments compare within a particular center?

Compare the CPAP and control means within each center.

$$\text{Effect in center } j: \quad \beta_1 + d_j$$

One treatment effect per center.

MARGINAL TREATMENT EFFECT

How do treatments compare on average across centers?

Average each treatment's mean equally across centers, then compare the averages.

$$\text{Average effect:} \quad \beta_1$$

With effects coding, the treatment deviations average to zero.

Estimated marginal means (EMMs) are model-based estimates of group means, averaged over other factors where appropriate. Comparing these means gives the treatment effects.

Estimating simple treatment effects with estimated marginal means

CODING AND EFFECTS

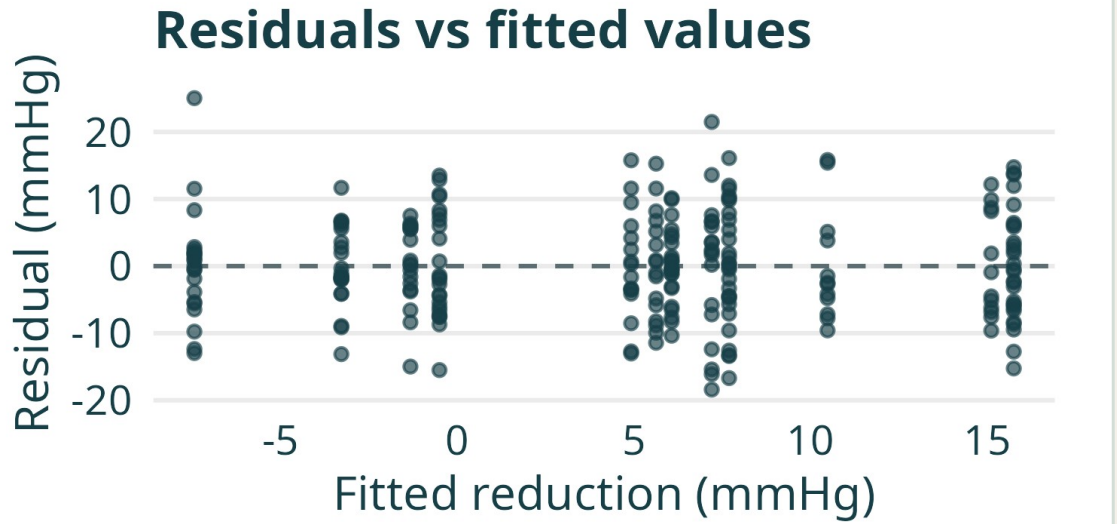
- Coding determines which effects appear directly as coefficients.
- Here, a center's effect combines the average effect with its treatment deviation.
- Predict both treatment means within each center, then compare them.

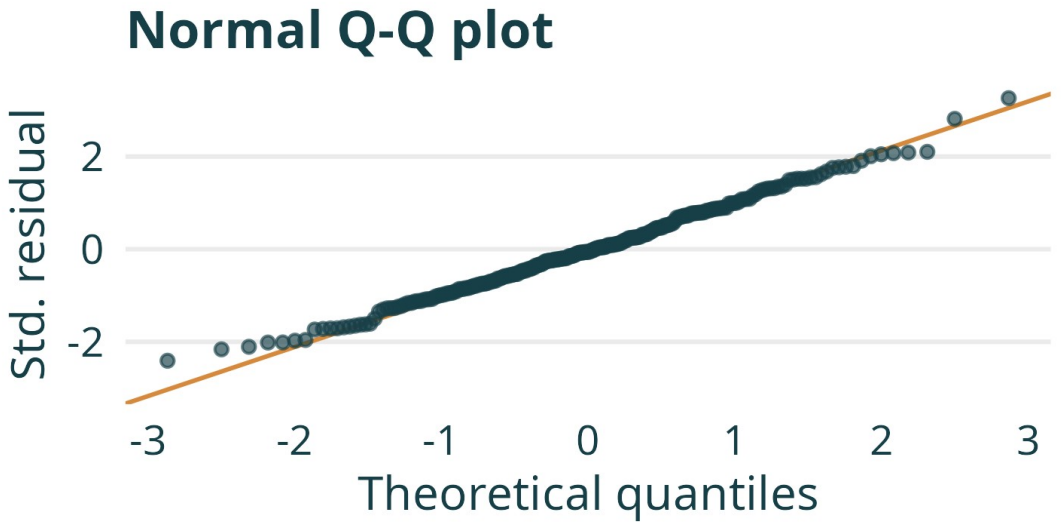
	Predicted means		Simple treatment effect		
Center	Control mean	CPAP mean	Difference	95% CI	p-value
A	10.5	15.1	4.6	-1.7 to 11.0	0.150
B	5.6	4.9	-0.7	-6.2 to 4.8	0.800
C	-1.3	7.2	8.5	3.4 to 13.7	0.001
D	-3.3	-7.4	-4.2	-8.8 to 0.5	0.081
E	-0.5	6.1	6.6	2.1 to 11.0	0.004
F	7.7	15.8	8.1	3.9 to 12.2	< .001

Means, differences and CIs in mmHg. Difference: CPAP minus control. Pointwise 95% CIs and unadjusted p-values.

Equivalent coding schemes give different coefficients, but the same fitted means and treatment comparisons.

Model diagnostics: fixed-effects interaction





WHAT WE USE THIS PLOT FOR

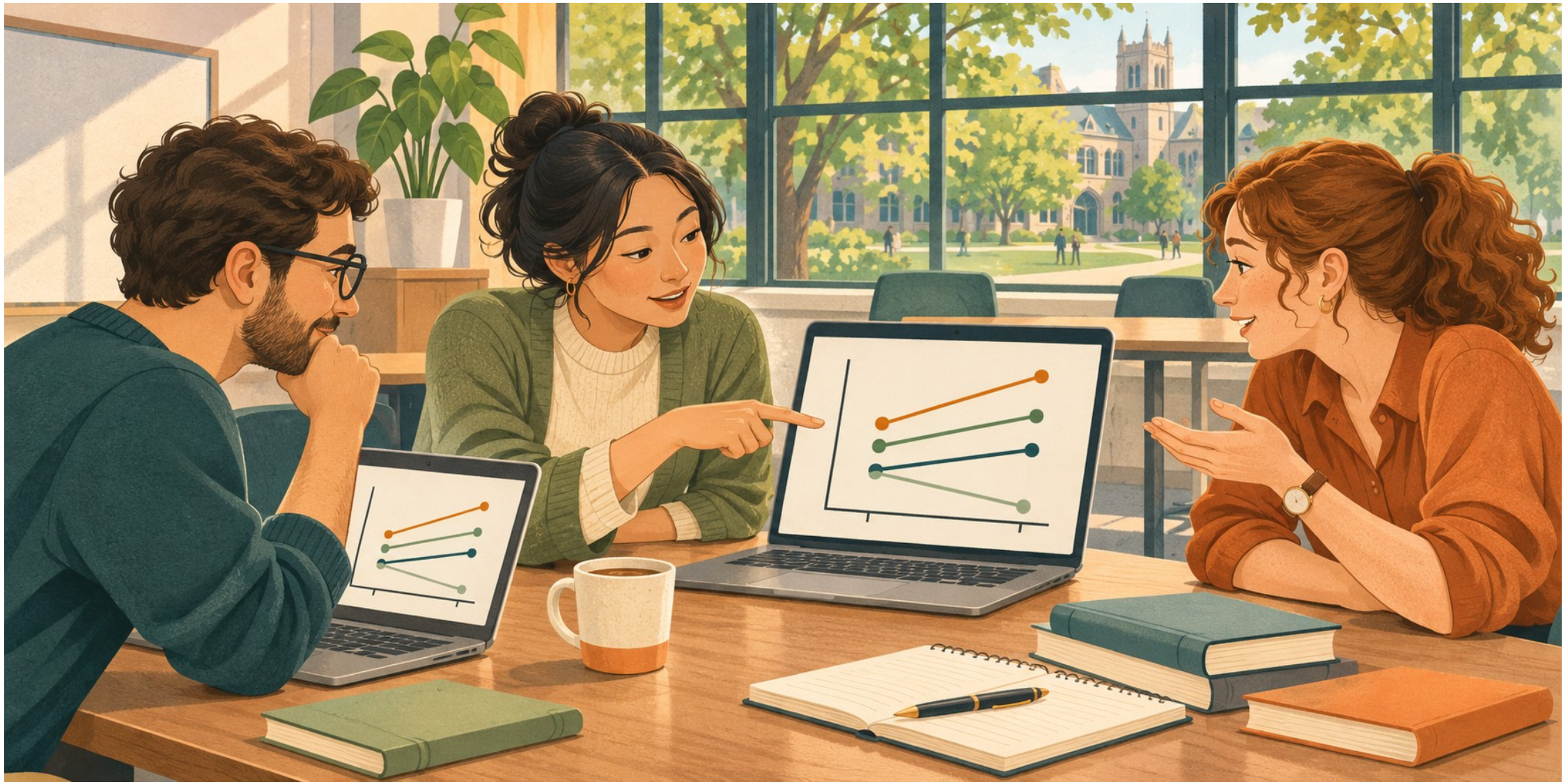
Assess whether residual spread is similar across fitted blood-pressure reductions.

WHAT WE USE THIS PLOT FOR

Assess whether the residual distribution is approximately Normal.

Residual spread is broadly similar across fitted values, and points lie close to the Normal reference line.

Time for Lab 3



Lab 3: what if we leave out the interaction?

Average effect across the five laboratories
Experimental minus standard, in percentage points of wound closure

Model	Estimate	Estimated SE	95% CI	p-value
Additive	9.28	1.81	5.69 to 12.88	< .001
With interaction	9.28	1.62	6.07 to 12.49	< .001

Same estimate: 10 plates per treatment in every laboratory make both models estimate the average of the five within-laboratory treatment differences.

Larger estimated SE: without interaction, differences in treatment effects remain in the residuals, increasing the estimated residual variation.

In this balanced study, omitting interaction leaves the average treatment estimate unchanged but increases its estimated uncertainty.

What if laboratory sizes differ?

WHAT THE MODELS ESTIMATE

- With our effects coding, the interaction model estimates the equally weighted average treatment effect, regardless of laboratory sizes.
- If laboratories have equal sizes, the additive model estimates this same average.
- If laboratory sizes differ, the additive model gives larger laboratories more weight.

HYPOTHETICAL EXAMPLE

Effects in percentage points

Laboratory	Plates	Treatment effect
Small	20	5
Large	80	15

Interaction model: 10 points

Additive model: 13 points

When treatment effects differ across laboratories, omitting interaction can change both the interpretation of the treatment coefficient and its estimated uncertainty.

PART 2

Treatment-by-center interaction: random effects

From the included centers to a population

THESE SIX HOSPITALS

Imagine repeating the study with new patients in the same six hospitals.

The centers' true treatment effects stay the same.

We estimate the average effect in these six hospitals.

A POPULATION OF CENTERS

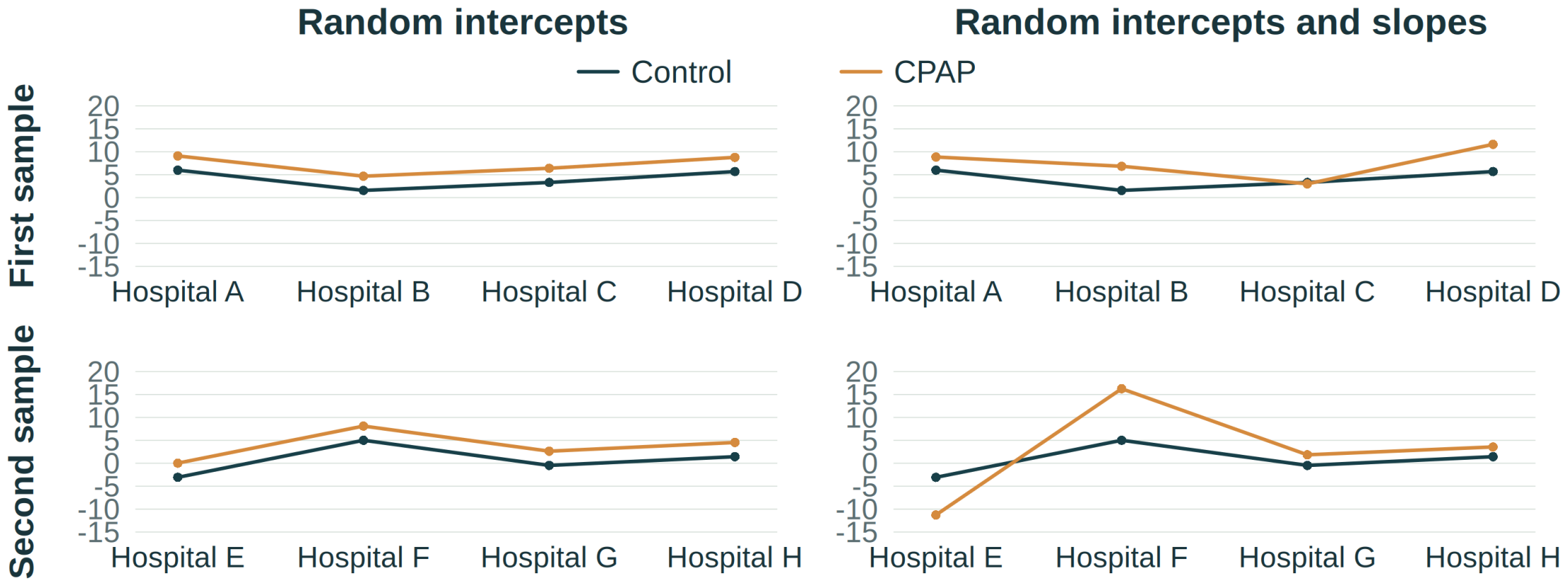
Imagine repeating the study with new centers from the same population.

The selected centers can have different true treatment effects.

We estimate the average effect across the population of centers.

When we generalize to a population of centers, uncertainty also depends on which centers enter the study.

What changes when we sample new centers?



Hypothetical true mean blood-pressure reductions (mmHg) across hospitals.

The population model stays the same, but each sample contains centers with different realized effects.

The center effects follow a population distribution

$$Y_{ij} = \beta_0 + c_j + (\beta_1 + d_j)T_{ij} + \varepsilon_{ij}$$

β_0 and β_1 are population averages. T_{ij} = 0 (control) or 1 (CPAP).

DISTRIBUTIONAL ASSUMPTIONS

$$\begin{pmatrix} c_j \\ d_j \end{pmatrix} \sim N\left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_c^2 & \sigma_{cd} \\ \sigma_{cd} & \sigma_d^2 \end{pmatrix}\right]$$

$$\varepsilon_{ij} \sim N(0, \sigma^2)$$

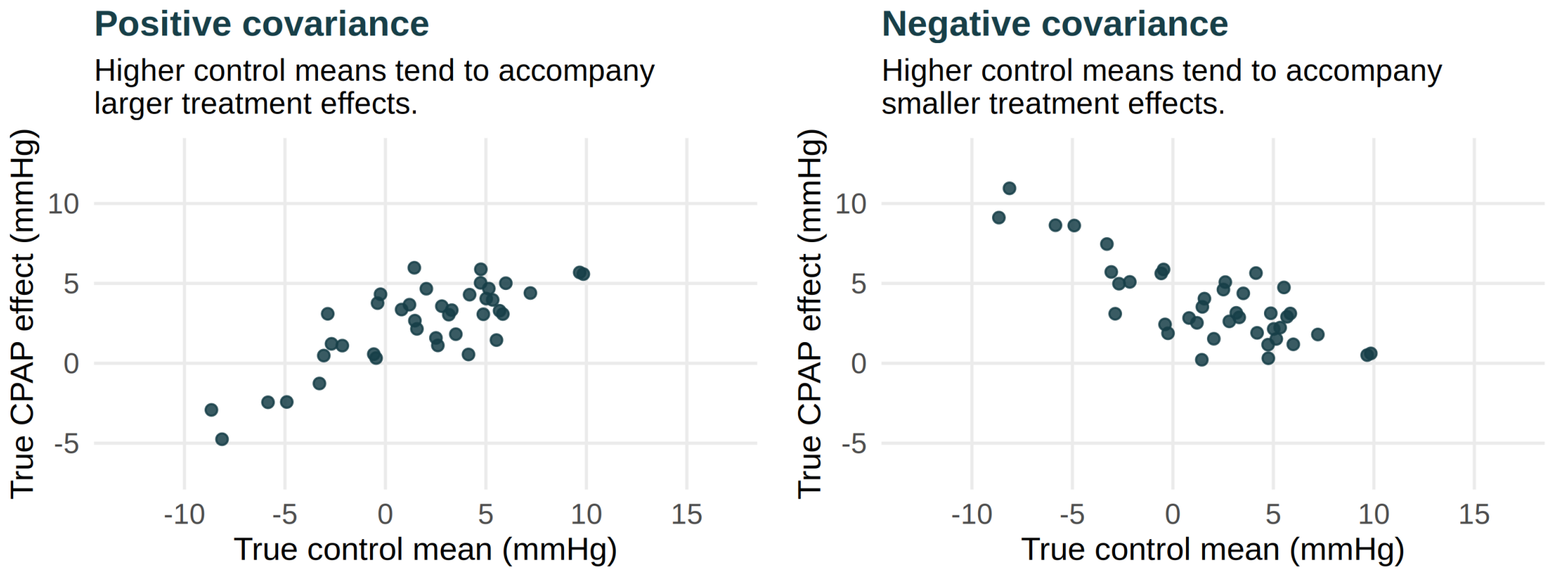
Effect pairs are independent across centers. Errors are mutually independent and independent of the center effects.

VARIANCES AND COVARIANCE

σ_c^2: variance of control means across hospitals.
σ_d^2: variance of treatment effects across centers.
σ_{cd}: covariance of control-mean and treatment-effect deviations.
σ^2: individual variance around the center-specific treatment mean.

Center deviations have population means of zero. Their average varies across samples of hospitals.

How do control outcomes and treatment effects vary together?



True control means and CPAP-minus-control effects (mmHg). Each point is one hypothetical hospital.

Covariance describes how a hospital's control outcome and treatment effect vary together.

CPAP study: estimated variance components

Results from the random-intercept and random-slope model fitted to the simulated CPAP data.

Component	Parameter	Estimate (mmHg²)
Control means across hospitals	σ_c^2	26.2
Treatment effects across centers	σ_d^2	20.5
Control-mean / treatment-effect covariance	σ_{cd}	11.2
Individual residual variation	σ^2	61.8

Estimated control-mean / treatment-effect correlation

$$\widehat{\rho}_{cd} = \frac{\widehat{\sigma}_{cd}}{\sqrt{\widehat{\sigma}_c^2 \widehat{\sigma}_d^2}} = \frac{11.2}{\sqrt{26.2 \times 20.5}} \approx 0.48$$

Hospitals with greater control-group reductions tend to have larger treatment effects.

Treatment-effect variance is comparable to control-mean variance. Both are smaller than individual residual variance.

The average treatment effect across the population of centers

The same 240 patients in six hospitals, fitted with random intercepts and treatment slopes.

Population parameter	Estimate	Estimated SE	95% CI	p-value
Average control-group reduction (β_0)	2.99	2.22	-2.7 to 8.7	0.237
Average treatment effect (β_1)	3.99	2.12	-1.4 to 9.4	0.117

Estimates, estimated SEs and confidence intervals are in mmHg.

The treatment coefficient estimates the average CPAP-minus-control effect across the population of centers.

Estimated average benefit: 4.0 mmHg. The 95% confidence interval includes zero.

Fixed and random effects: estimates and uncertainty

Hospital sizes differ, with equal treatment allocation within every hospital.

Model	Estimated treatment effect (mmHg)	Estimated SE (mmHg)	p-value
Fixed interaction	3.82	1.06	< .001
Random intercept and slope	3.99	2.12	0.117

Fixed interaction: equally weighted average for these six hospitals. Uncertainty comes from sampling patients.

Random intercept and slope: population-average effect. Uncertainty also reflects which hospitals entered the study.

Unequal hospital sizes allow different estimates. Between-hospital treatment variation adds uncertainty about the population average.

What if we omit the random treatment slope?

Back to the CPAP study: both models target the population-average treatment effect.

Model	Estimated treatment effect (mmHg)	Estimated SE (mmHg)	p-value
Random intercept only	4.08	1.05	< .001
Random intercept and slope	3.99	2.12	0.117

Random intercept only: assumes the same treatment effect in every center.

Random intercept and slope: allows treatment effects to vary across centers.

When treatment effects vary across centers, a random-intercept-only model can underestimate uncertainty about the population-average effect.

Time for Lab 4



Variance decomposition depends on the random-effects structure

WEEK 1: RANDOM INTERCEPT

The same center contribution applies under both treatments.

Outcome variance separates into a center component and an individual component.

This gives one common within-center correlation: the ICC.

WEEK 2: RANDOM INTERCEPT AND SLOPE

The center contribution depends on treatment.

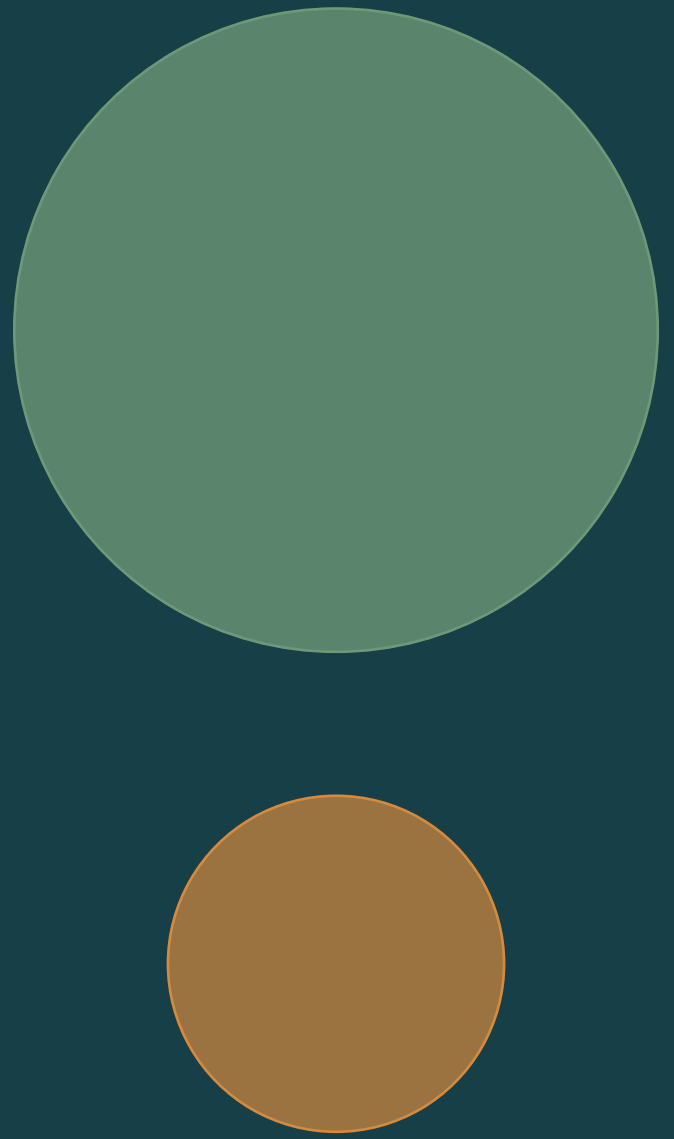
Outcome variance depends on treatment and combines the relevant variances and covariance.

Within-center correlation depends on the treatments received by the two patients.

Variance decomposition still applies, but there is no longer one common variance breakdown or a single ICC for all patients.

PART 3

Crossed random effects



What if there is more than one grouping variable?

So far: one grouping variable, center. What changes with two or more grouping variables?

NESTED GROUPING VARIABLES

Each group at one level belongs to exactly one group at the next level.

Patients belong to wards, and each ward belongs to one hospital.

Grouping variables: ward and hospital.

CROSSED GROUPING VARIABLES

Neither grouping variable is contained within the other.

Each patient is assessed by several radiologists, and each radiologist assesses several patients.

Grouping variables: patient and radiologist.

Next: crossed grouping variables, illustrated by patients and radiologists.
Week 3: nested grouping variables, such as wards within hospitals.

Case study: agreement between radiologists

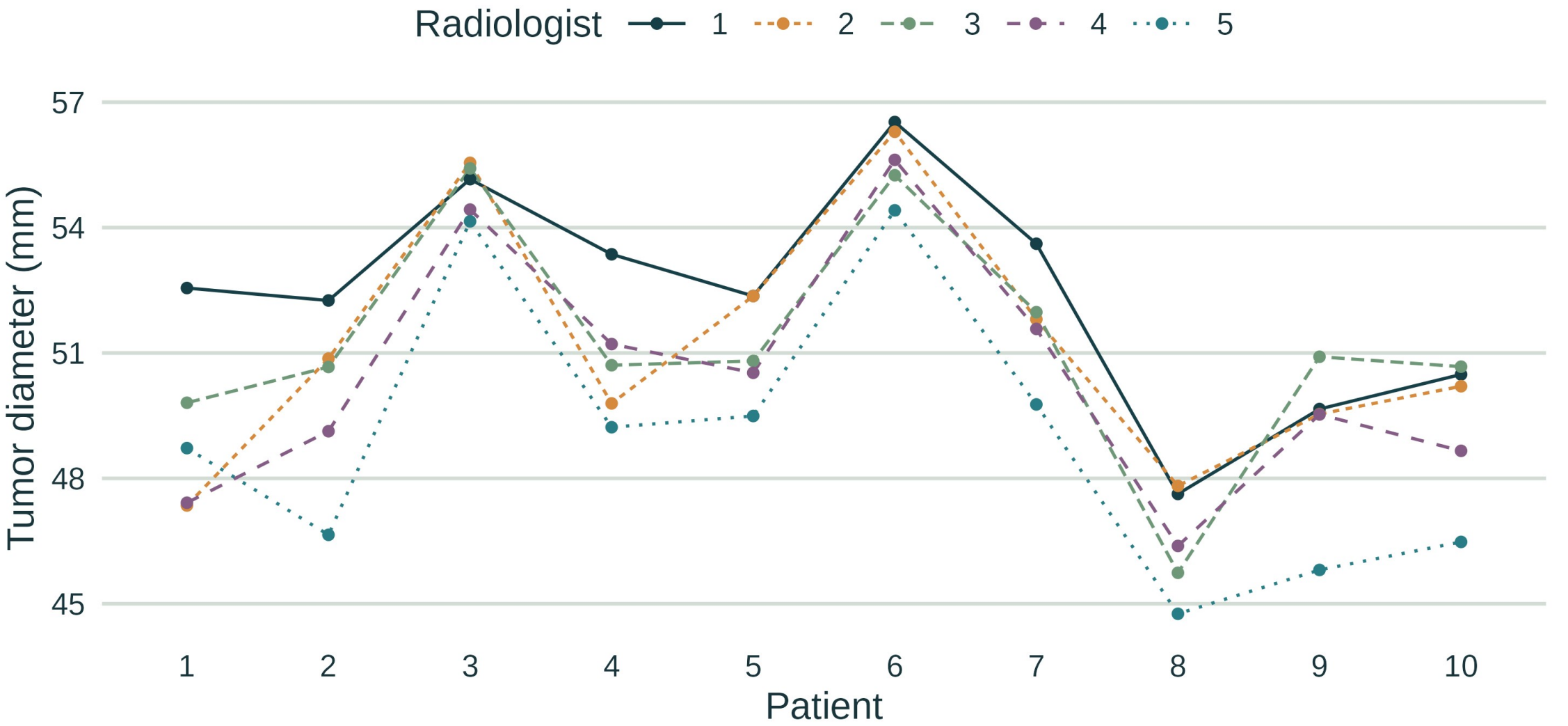
MEASURING TUMOR DIAMETER

- Simulated study: CT scans from 10 patients assessed independently by five radiologists.
- Every radiologist measures the maximum tumor diameter on every scan.
- Outcome: tumor diameter in millimeters, giving 50 measurements.
- We want to generalize beyond these patients and radiologists.

Do different radiologists obtain similar measurements for the same patients?



Exploring the radiologists' measurements



A model with two crossed grouping variables

$$Y_{ij} = \mu + u_i + v_j + \varepsilon_{ij}$$

Y_{ij} is the measurement for patient i made by radiologist j.

WHAT CONTRIBUTES TO A MEASUREMENT?

μ: overall mean tumor diameter.

u_i: patient i's deviation from the mean.

v_j: radiologist j's tendency to measure higher or lower.

ε_{ij}: remaining measurement variation.

TWO RANDOM INTERCEPTS

$$u_i \sim N(0, \sigma^2_{\text{patient}})$$

$$v_j \sim N(0, \sigma^2_{\text{radiologist}})$$

$$\varepsilon_{ij} \sim N(0, \sigma^2)$$

Patient effects, radiologist effects and residual errors are mutually independent.

Measurements are related when they share the same patient or the same radiologist.

Where does measurement variation come from?

Estimated variance components from the crossed model fitted to the radiologist study.

Source of variation	Estimate (mm^2)	Interpretation
Patient	7.44	Differences between patients
Radiologist	1.47	Systematic radiologist differences
Residual	1.00	Remaining measurement variation

$$\widehat{\mathrm{ICC}} = \frac{7.44}{7.44 + 1.47 + 1.00} \approx 0.75$$

Intraclass correlation coefficient (ICC): absolute agreement for a single measurement.

About 75% of the estimated total variation is between patients. Radiologist differences and residual variation reduce agreement.

The End

Questions



